



ABMS SESSION

Entrusting AI in Assessment and Certification

What we can learn from how we already entrust trainees

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Disclosures

Dr. Winn and Dr. Chiel receive funding from the ABP

Dr. Chiel receives funding from ABMS

Objectives

By the end of this session, we hope you will be able to:

- 1 Apply the entrustment frameworks we already use for trainees, including trustworthiness dimensions and supervision levels, to AI tools
- 2 Use the interplay of stakes, supervision, and trustworthiness, within its context, to decide how much to entrust an AI tool with a given task
- 3 Describe how one team designed, validated, and piloted an AI tool to assist with processing narrative comments for CCCs
- 4 Analyze potential uses of AI in board certification through the same entrustment lens

What is AI — and what is a large language model?

ARTIFICIAL INTELLIGENCE *tasks that normally take human intelligence*

MACHINE LEARNING *learns patterns from data*

LARGE LANGUAGE MODELS *ChatGPT · OpenEvidence*

Ingests enormous amounts of text and predicts the next word — again and again. **It does not understand. It predicts.**

Non-generative use
gives you information

Generative use
creates new content

Most of today's examples involve large language models — not predictive machine learning.

What a large language model actually does

Training knowledge

Patterns from billions of words — baked in; unchanging.

Your data

What you provide — temporary; it does not train the model.

Your instructions

The instruction — same model + different instructions = different output.



OUTPUT — training knowledge applied to your data, guided by your instructions.

Data privacy is its own major consideration in designing and using these tools — a topic of its own, not addressed today.

Three things the soccer player teaches us



*A model, like a player,
is its training*

Different training, different player

Same play call, different execution

It can't fully explain the play

Two judgments we already make every day



1. Is this *trainee* trustworthy?

- Do they have the knowledge and skill?
- Do they know their limits?
- Do they report honestly?
- Are they dependable, every time?



2. How much do I/we entrust *them*?

Judgment 1 feeds in — plus the circumstance.

- Demonstrated trustworthiness...
- ...plus the task and the stakes

Two judgments we already make every day



1. Is this *trainee* AI tool trustworthy?

A judgment about the agent.

- Do they have the knowledge and skill?
- Do they know their limits?
- Do they report honestly?
- Are they dependable, every time?



2. How much do we entrust *them* the AI tool?

Judgment 1 feeds in — plus the circumstance.

- Demonstrated trustworthiness...
- ...plus the task and the stakes

Today's exercise: replace *trainee* AI tool — and notice that the questions don't change.

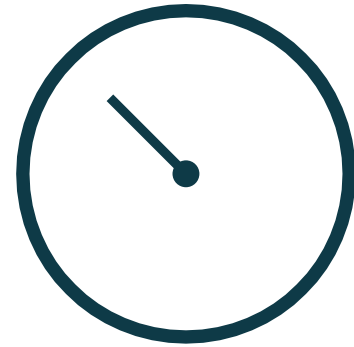
Stakes, supervision, and trustworthiness are interrelated



STAKES



TRUSTWORTHINESS

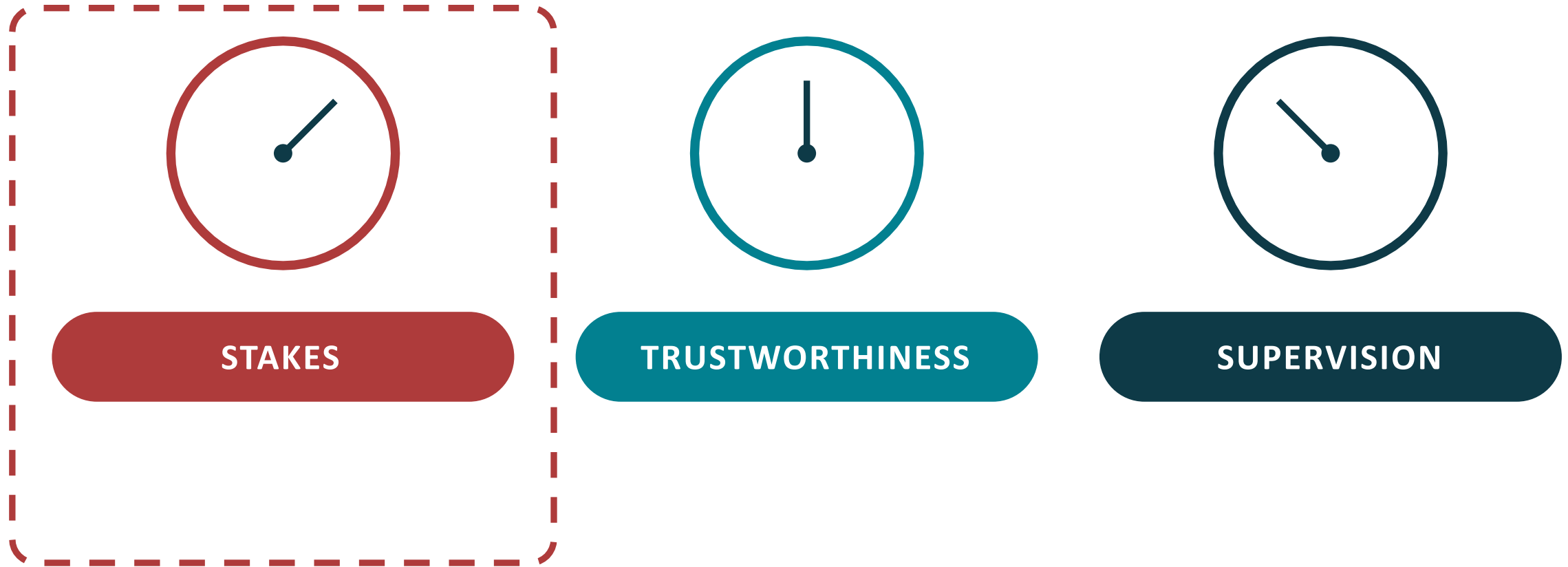


SUPERVISION

Turn one dial, and the others must move — we'll take each in turn.

Adapted from the five factors of entrustment (Hauer KE, ten Cate O, et al. Adv Health Sci Educ 2014) — not yet published; a frame others can use.

Stakes, supervision, and trustworthiness are interrelated



Dial 1 · Stakes: what happens if the output is wrong?

Stakes are a property of the task and its use — not of the tool.

WHAT MOVES THIS DIAL

- Severity
- Reversibility
- Who bears the consequence

STAKES RISE

Drafting your manuscript

Writing an email to two colleagues

Generating notes for yourself

Stakes, supervision, and trustworthiness are interrelated



You've seen the headlines

Lawyers sanctioned after AI invents six court cases

Chatbot rolled back for telling users what they want to hear

AI cites studies that don't exist — in published manuscripts

No one can explain why the model said that

Confidently wrong, again: chatbot flunks the basics

Can medicine trust AI?

...so what does “trustworthy” actually mean?

Trustworthiness: four dimensions — one word swapped



Knowledge & skill

Does the *trainee* **AI tool** have the knowledge and skill?

- ✓ grounded in pediatric evidence
- ✗ trained mostly on adults



Discernment of limitations

Does the *trainee* **AI tool** know when it's out of its depth?

- ✓ admits “I can’t answer that”
- ✗ fabricates confidently



Truthfulness

Does the *trainee* **AI tool** report honestly?

- ✓ claims traceable to source
- ✗ invents citations



Conscientiousness

Does the *trainee* **AI tool** follow through, every time?

- ✓ same rigor, every run
- ✗ skims and cuts corners

Not all AI tools are alike: same question, two agents

A general-purpose chatbot

e.g., ChatGPT

the broad internet:
wide, uneven

bluffs when wrong

unsourced — verifying
is your job

follow-through unproven

Knowledge & skill

Discernment

Truthfulness

Conscientiousness

A clinically grounded tool

e.g., OpenEvidence

the medical literature:
deeper

says when evidence is thin

cites retrievable sources

also unproven

Different trustworthiness → different supervision, for the very same task.

Stakes, supervision, and trustworthiness are interrelated



STAKES



TRUSTWORTHINESS



SUPERVISION



Dial 3 - Supervision

Who is in the loop?

Trainee



AI tool



Supervision: built in, then tailored

Built in

the same for everyone



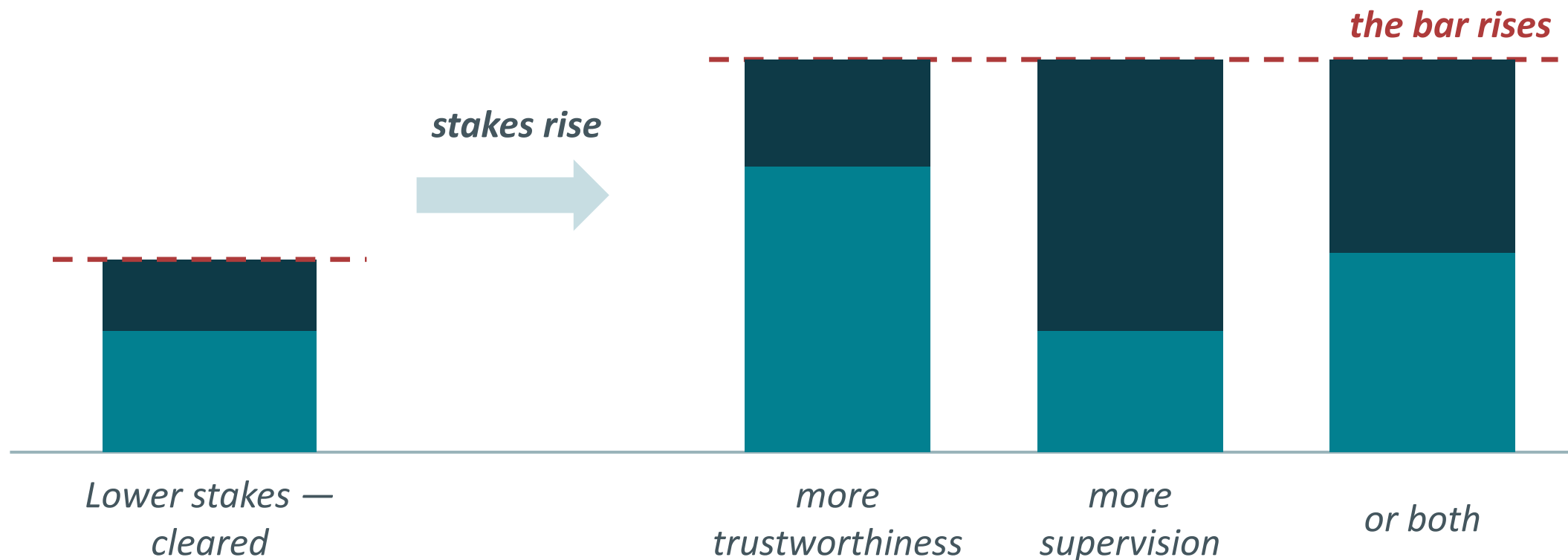
Tailored

more hoops as stakes rise
fewer hoops as trust is earned



As stakes rise, the bar rises — either dial can meet it

--- the bar (stakes) ■ trustworthiness demonstrated ■ supervision, added on top



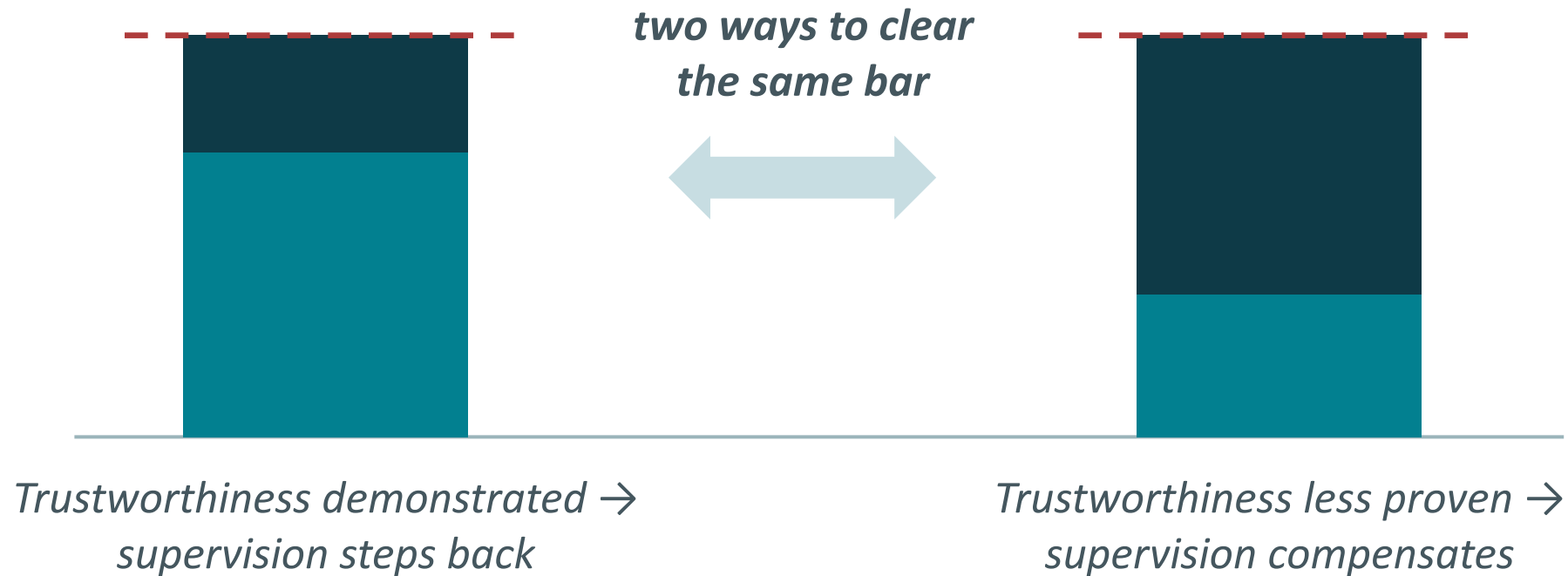
Either dial can close the gap: more demonstrated trustworthiness, more supervision — or both.

Within the same stakes, trustworthiness and supervision trade off

--- same stakes, same bar

■ trustworthiness demonstrated

■ supervision, added on top



Supervision needed = the bar set by the stakes – the trustworthiness proven.

Context shapes every entrustment decision



Context changes what each dial must carry — but patient safety is a floor it can never lower.

Trust evolves — in both directions



Trainees AI tools earn autonomy

Like a resident progressing:

- Validity evidence accumulates
- Supervision loosens step by step
- Earned by evidence



New *interns* AI tools keep arriving

New interns every July; new AI every quarter:

- New tools start at the bottom

Today's entrustment decisions are provisional by design.

Bringing it back to ABMS.....





WORKED EXAMPLE

AI to assist with processing narrative comments by a Clinical Competency Committee

Laura Chiel & Ting Wang

From a dearth of narrative comments to an overwhelming abundance

THEN

A dearth of narrative data

- Sparse, end-of-rotation comments
- Generic: “great to work with”
- Decisions on thin signal



NOW · WBA AT SCALE

An abundance that overwhelms

- Frequent, EPA-anchored assessments
- Hundreds of comments per resident
- More than a CCC can read — the signal is buried

Narrative comments provide critically important information — supplementing ratings for more accurate entrustment assessment

Who has used or heard of AI for narrative comments in CCC work?



I've used it myself

I know someone who has

I've only heard about it

What would make you NOT trust AI to assist with processing narrative comments?



Call out your concerns — short phrases are perfect.

Let's categorize what you said: four buckets, plus one

As the concerns come out, notice where each lands:

1 Knowledge & skill

"It will synthesize inaccurately"

2 Discernment of limits

"It won't say when data are thin"

3 Truthfulness

"It will make things up"

4 Conscientiousness

"Different answer every run"

5 **...or it's not about the tool at all.** Who uses it · how it's used · the risk · the setting — those set the supervision level.

Kennedy's dimensions for a ~~trainee~~ AI tool — you already knew how to think about this.

Two ways to use AI with assessment data

Same LLM — the difference is the ask. This distinction is exactly why we chose organization first.

Organize — keep the original text

- Original comments stay intact; the model sorts, groups, extracts, flags
- Every output traceable back to a source comment
- Lower risk, more transparent, more auditable
- e.g., sort by milestone; group similar feedback; flag safety concerns

Synthesize — write new text

- New text is produced — the model writes in its own words
- The committee reads AI-generated text, not original faculty language
- Higher risk; verification is harder (no gold standard)
- e.g., synthesized summaries; drafted progress reports; discussion points

Our first entrustment decision: organize, don't synthesize

STAKES

Program directors judged synthesis too risky, for now — organizing delivers value at low stakes.

TRUSTWORTHINESS

No gold standard exists for the “right” synthesis. Organization is checkable.

SUPERVISION

Synthesis could quietly end the reading — organization keeps humans reading.

Design starts from the outcome we want

Competency-based medical education makes the same move: start from the outcome — the needs of patients — and work backward.



The outcome

A more accurate, comprehensive picture of each trainee — enabling defensible decisions and tailored learning plans, in service of learners and their patients.



What program directors asked for

Organization built on the constructs programs already use — EPAs and milestones — with source comments one click away. The tool prepares; the committee decides.

Feedback Dashboard: Classification Design Principles

Every narrative comment is tagged by two complementary sources — deterministic rules and a zero-shot, multi-agent language-model pipeline

EPA 10

Rule-based · *deterministic*

The assessment form's **activity + setting** map to a parent EPA through the **SIMPL crosswalk** — no model judgment involved.

Deterministic and auditable: the anchor EPA is always present and never overridden by the model.

Setting rule: any Nursery/Delivery comment also receives **EPA 5** (newborn care).

ICS1

Text-based · *language model*

The model reads the **comment narrative itself**, adding EPAs beyond the anchor and assigning all ACGME Milestones.

Evidence gate: a tag is assigned only when the text substantively supports it.

Every tag carries a “**why**” **rationale** quoting the comment, so reviewers can verify the reasoning.

0

Zero-shot, agentic

No task-specific training data: prompts are generated from a single-source EPA / Milestone taxonomy, keeping definitions consistent.

Three specialized agents process each comment — summarize, then classify.

Learned rules and examples are injected as feedback accrues — **no retraining**.

Per-comment multi-agent pipeline

Narrative comment
+ form activity, setting,
question



Agent A — Summarizer
distills the comment into a
theme summary



Agent B — EPA classifier

rule-based anchor + text-based EPAs

Agent C — ACGME classifier

text-based Milestone subcompetencies

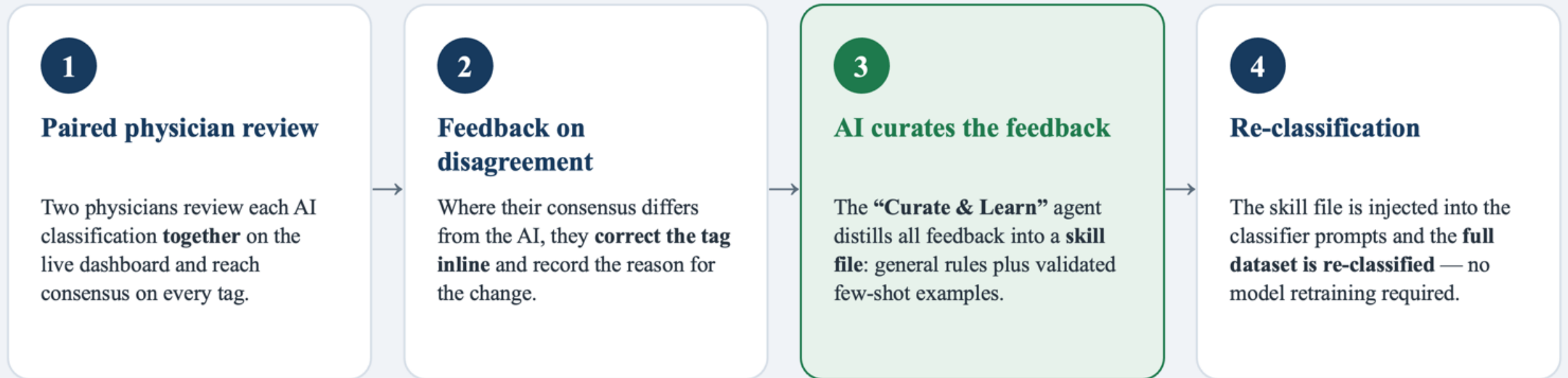


Tagged comment → dashboard

EPA + Milestone badges, each with a “why” rationale

Validation Plan: Paired Physician Review with an AI Learning Loop

Human consensus is the gold standard; the dashboard learns from every disagreement and re-classifies until agreement converges



🔄 **Iterate:** the cycle repeats — review → feedback → skill file → re-classification — until physician–AI agreement converges

Why this design: physician consensus provides the validity evidence; disagreements become training signal instead of errors; and because learning lives in a transparent, human-readable skill file — not model weights — every adjustment is reviewable and reversible.

ACGME Milestone Summary

TOTAL COMMENTS

571

All narrative comments

SUCCESSFULLY CLASSIFIED

543

95% of total

TOTAL CLASSIFICATIONS

1657

2.9 avg per comment

UNCLASSIFIED

28

28 placeholder + 0 no match

General Pediatrics EPA Summary

TOTAL COMMENTS

571

All narrative comments

SUCCESSFULLY CLASSIFIED

491

86% of total

TOTAL CLASSIFICATIONS

784

1.4 avg per comment

UNCLASSIFIED

80

80 placeholder + 0 no match

All ACGME Milestone Categories

PC1	History	105	PC2	Physical Exam	60
PC3	Organize and Prioritize Patient Care	152	PC4	Clinical Reasoning	178
PC5	Filter to PC3 Management	232	MK1	Clinical Knowledge	99
MK2	Diagnostic Evaluation	56	SBP1	Patient Safety	7
SBP2	Quality Improvement	2	SBP3	Coordination of Care	69
SBP4	Transitions in Care	30	SBP5	Population and Community Health	11
SBP6	Physician Role in Health Care Systems	12	ICS1	Patient- and Family-Centered Communication	199
ICS2	Interprofessional and Team Communication	173	ICS3	Communication within Health Care Systems	72
PROF	Professionalism	84	PBL11	Evidence-Based and Informed Practice	43
PBL12	Reflective Practice and Commitment to Personal Growth	73			

All 12 General Pediatrics EPAs

EPA 1	Preventative Primary Care for Children of All Ages	69	EPA 2	Comprehensive Care for Complex, Chronic, or Special Health Care Needs	108
EPA 3	Managing Common Acute Diagnoses	154	EPA 4	Common Behavior and Mental Health Concerns	35
EPA 5	Caring for the Newborn Prior to Discharge	54	EPA 6	Recognizing Need for Subspecialty Care and Seeking Referral	55
EPA 7	Recognizing the Severely Ill Patient and Mobilizing Resources	48	EPA 8	Executing Clinical Handovers	57
EPA 9	Common Procedures of the General Pediatrician	10	EPA 10	Leading Interprofessional Teams for Family-Centered Care	100
EPA 11	Promoting Equitable Care to Address Racism and Health Inequities	27	EPA 12	Utilizing Technology and Information Sources	67

All Individuals ▾

Search in comments...

All EPAs ▾

All ACGME ▾

All Settings ▾

All PGY ▾

All Feedback Status ▾

Laura

Curate & Learn

***** did a lot of spontaneous teaching for our medical student and even the MTAs on rounds , explaining things about patients they may have missed and really setting the medical student up for success on family centered rounds. I can see her passion for teaching and she is great at it! I'm so glad she is taking the initiative to help the other learners on the team and I know it makes the rotation so much better for them.

Feedback

LEAD a Team and/or SUPERVISE others

Hospitalist/GP Inpatient

On inpatient rounds, the resident engaged in spontaneous teaching for the medical student and MTAs, explaining aspects of patient care they may have missed and preparing the medical student for family-centered rounds. The feedback highlights initiative in supporting other learners on the team.

EPA 10
► why

MK1
► why

ICS2

PGY-1

edit

*** did a really good job of seeking out feedback from me throughout his rotation. In addition to asking me specifically for feedback, he was interested in improving his assessments and his notes. And he incorporated my feedback into his notes subsequent days.

Commitment to Growth and Learning

Single Free Text Question- COMMITMENT TO GROWTH AND LEARNING

Hospitalist/GP Inpatient

During an inpatient hospitalist rotation, the resident actively sought feedback from the evaluator throughout the rotation and expressed interest in improving assessments and notes. He then incorporated that feedback into his notes on subsequent days.

none

PBL12
► why

ICS3

PGY-1

edit

Next: study it in vivo — 3 pilot sites

Ting asked: does the tool do what it was intended to do?

The pilot asks: does it function as intended within programs — without unintended consequences?



The setting

- 3 programs, different contexts
- Real CCCs, real decisions

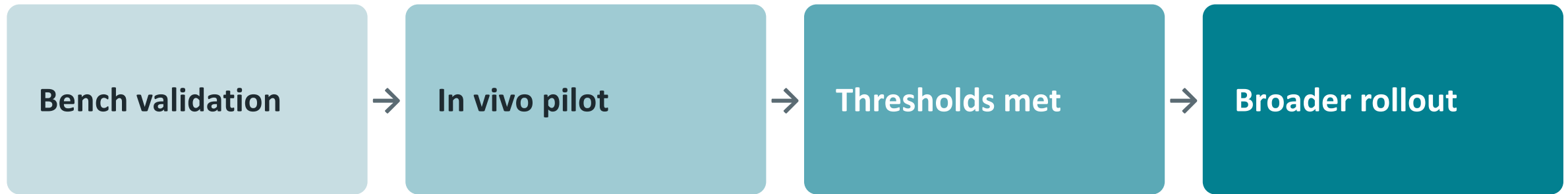


What we'll study

- Fidelity in the wild — real, messy data
- Workload — does it help CCCs?
- Effect on discussion and decisions
- Use as intended — does review happen?
- Equity — consistent across residents

Not the tool alone — the interaction of tool, users, and workflow together.

Scale only when evidence moves it up the ladder



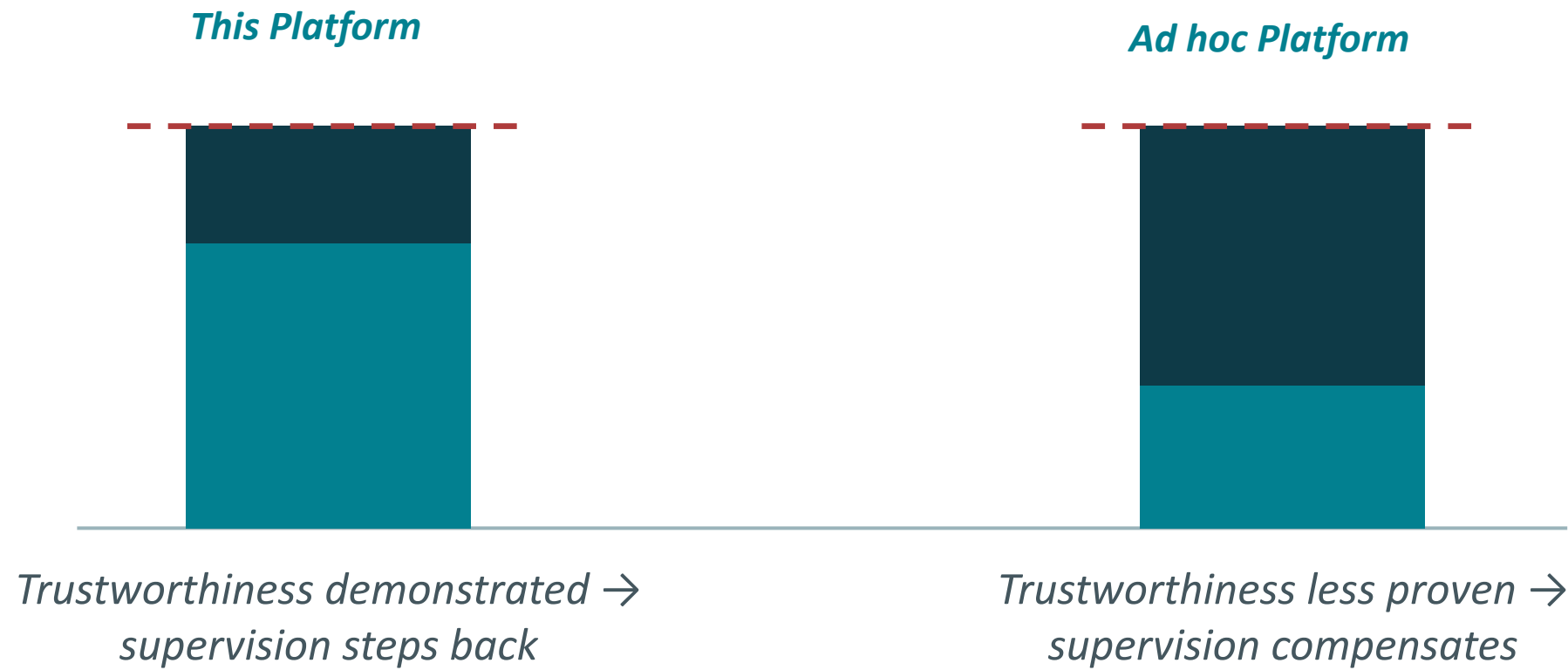
Exactly how residents progress:

- Supervision loosens one notch at a time, on evidence — never on a demo
- Thresholds set in advance
- Monitoring continues after scaling — drift, misuse, fairness
- Any red flag moves it back down — demotion is part of the system
- Entrustment parameters are set locally — they vary by program and context

Then we can feel good about trusting it in broader settings.

Different trustworthiness, different supervision needed

--- same stakes, same bar ■ trustworthiness demonstrated ■ supervision needed





RAISING THE STAKES

How does AI fit into certification processes?

*There are many possibilities:
item development · standard setting · scoring · portfolio review · data
synthesis · decision support*

David A. Turner, MD
Vice President, Innovation and Competency-Based Education
American Board of Pediatrics

Certification is itself an entrustment decision

The supervisor entrusts a trainee with today's task, for this patient

The program & CCC entrust the graduate with unsupervised practice

The board certifies — the profession's promise to the public

Same decision, rising stakes and reach: when is the individual ready?

When and how we should 'entrust' AI in certification processes is a decision about one of the most consequential decisions.

A Concrete Example - AI *drafts* exam items

STAKES

Medium – As long as the focus is on *drafting* items for human review approval

SUPERVISION

High - Built in already through existing systems

TRUSTWORTHINESS

AI can be trusted to *contribute to* the development of accurate, unbiased items

Verdict: entrustable today — AI drafts, humans review every item

Concrete Example 2 – Fully Automated AI driven item generation

STAKES

High – But likely can be mitigated with human input at certain points in the process

SUPERVISION

Needs to be adjusted to balance AI capabilities for different parts of the process with the need for human review/input

TRUSTWORTHINESS

AI likely can be trusted, but many systems are not yet equipped for this degree of integration

Verdict: AI is NOT entrustable today — but may be tomorrow...

Stakes, supervision, and trustworthiness are interrelated



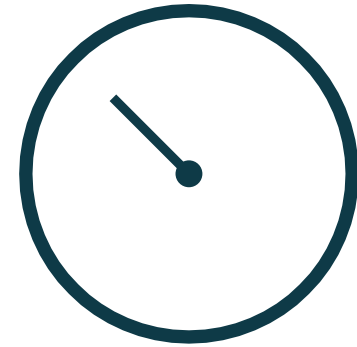
STAKES

*consequence of error —
set by the task*



TRUSTWORTHINESS

*what the tool has
demonstrated*



SUPERVISION

*the human review planned
to meet the need*

Turn one dial, and the others must move

Adapted from the five factors of entrustment (Hauer KE, ten Cate O, et al. Adv Health Sci Educ 2014) — not yet published; a frame others can use.

All ACGME Milestone Categories

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PC3	Organize and Prioritize Patient Care	117	PC4	Clinical Reasoning	189
PC5	Patient Management	219	WK1	Clinical Knowledge	85

All Individuals 0

Search in comments...

All EPAs 0

All ACGME 0

Thank you

KEY REFERENCES

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Questions & discussion